Second and Higher Derivatives
Answer Key

AP Calculus AB/BC   Implicit, Inverse, and Higher-Order Differentiation

Quick Answers

1. $20x^3 - 18x$
2. $48x$
3. $-9\sin(3x)$
4. $\left(x^2 + 4x + 2\right)e^x$
5. $6t - 12$, 2
6. $6x - \frac{1}{x^2}$, 11.75
7. $-\frac{25}{(25-x^2)^{\frac{3}{2}}}$, -0.390625
8. $12t^2 - 24t$, 0, 2

Curriculum

Level: AP Calculus AB/BC   Implicit, Inverse, and Higher-Order Differentiation
FUN-3.D–FUN-3.E – *problems 1, 2, 3, 4, 5, 6, 7, 8*
Difficulty 1–5 of 5 across 13 tagged checks.

1. $f(x) = x^5 - 3x^3 + 2x$; find $f''(x)$.
Power rule, one rung at a time:

$$f'(x) = 5x^4 - 9x^2 + 2 \qquad \Longrightarrow \qquad f''(x) = 20x^3 - 18x$$

$$\boxed{f''(x) = 20x^3 - 18x}$$

Check the pattern: each rung drops the degree by one, so a quintic has a cubic second derivative ✓

2. $g(x) = 2x^4 - x^2 + 7$; find $g'''(x)$.
Three rungs. The constant 7 dies on the first rung and never returns:

$$g'(x) = 8x^3 - 2x \qquad g''(x) = 24x^2 - 2 \qquad g'''(x) = 48x$$

$$\boxed{g'''(x) = 48x}$$

Note for the grader: a fourth derivative would be the constant 48, and every derivative after that is 0 — the degree, 4, is exactly how many rungs it takes to reach a constant.

3. $f(x) = \sin(3x)$; find $f''(x)$.
Chain rule on each rung; the inner derivative is 3 both times:

$$f'(x) = 3\cos(3x) \qquad \Longrightarrow \qquad f''(x) = 3 \cdot \big(-\sin(3x)\big) \cdot 3 = -9\sin(3x)$$

$$\boxed{f''(x) = -9\sin(3x)}$$

Common error: writing $-\sin(3x)$, which drops the inner derivative on the second rung. The factor of 3 appears once per differentiation, so the second derivative carries 3^2.

4. $h(x) = x^2 e^x$; find $h''(x)$.
Product rule twice, simplifying between rungs:

$$h'(x) = 2xe^x + x^2e^x = e^x(x^2 + 2x)$$
$$h''(x) = e^x(x^2 + 2x) + e^x(2x + 2) = e^x(x^2 + 4x + 2)$$

$$\boxed{h''(x) = (x^2 + 4x + 2)e^x}$$

Why factoring first pays: differentiating the unsimplified $2xe^x + x^2e^x$ needs the product rule on two terms instead of one.

5. Drone: $s(t) = t^3 - 6t^2 + 9t$ metres.
(a) Acceleration. Velocity is the first rung, acceleration the second:

$$v(t) = s'(t) = 3t^2 - 12t + 9 \qquad \Longrightarrow \qquad a(t) = s''(t) = 6t - 12$$

in metres per second squared. $\boxed{a(t) = 6t - 12}$
(b) Zero acceleration. Solve $6t - 12 = 0$, giving $t = 2$ seconds — inside the stated window $0 \le t \le 5$. $\boxed{t = 2}$
Interpretation to look for: before $t = 2$ the acceleration is negative and the drone is slowing its rise; after $t = 2$ it is positive. That instant is the inflection point of the height graph.

6. $f(x) = \ln x + x^3$, $x > 0$.
(a) Second derivative. The derivative of $\ln x$ is x^{-1}, whose derivative is $-x^{-2}$:

$$f'(x) = \frac{1}{x} + 3x^2 \qquad \Longrightarrow \qquad f''(x) = -\frac{1}{x^2} + 6x$$

$$\boxed{f''(x) = -1/x^2 + 6x}$$

(b) Value and concavity. At $x = 2$: $f''(2) = -\frac{1}{4} + 12 = 11.75$. $\boxed{f''(2) = 11.75}$
Because $f''(2) > 0$, the graph of f is concave up at $x = 2$. *Common error:* differentiating x^{-1} to x^{-2} without the sign change, which flips the concavity conclusion.

7. Circle $x^2 + y^2 = 25$; find $\dfrac{d^2y}{dx^2}$ and its value at $(3, 4)$.

(a) First rung. Differentiate both sides with respect to x, treating y as a function of x:

$$2x + 2y\frac{dy}{dx} = 0 \qquad \Longrightarrow \qquad \frac{dy}{dx} = -\frac{x}{y}$$

Second rung. Quotient rule on $-x/y$, remembering that y is a function of x:

$$\frac{d^2y}{dx^2} = -\frac{y - x\dfrac{dy}{dx}}{y^2} = -\frac{y - x\left(-\dfrac{x}{y}\right)}{y^2} = -\frac{y^2 + x^2}{y^3}$$

Now substitute the original equation, $x^2 + y^2 = 25$:

$$\boxed{\frac{d^2y}{dx^2} = \frac{-25}{(25 - x^2)^{3/2}}}$$

which is the same as $-25/y^3$ on the upper arc, where $y = \sqrt{25 - x^2}$.

(b) At $(3, 4)$. Substituting $y = 4$: $-25/4^3 = -25/64$. $\boxed{-\dfrac{25}{64}}$

Sanity check: the point $(3, 4)$ is on the upper arc, which is concave down, and the value is negative ✓

8. Particle: $s(t) = t^4 - 4t^3 + 12$ metres.

(a) Acceleration.

$$v(t) = s'(t) = 4t^3 - 12t^2 \qquad \Longrightarrow \qquad a(t) = s''(t) = 12t^2 - 24t$$

in metres per second squared. $\boxed{a(t) = 12t^2 - 24t}$

(b) Zero acceleration. Factor: $12t^2 - 24t = 12t(t - 2)$, so $a(t) = 0$ at $t = 0$ and $t = 2$ seconds, both in the domain $t \geq 0$. $\boxed{t = 0,\ t = 2}$

(c) Concavity (manual review). The factored form $12t(t - 2)$ is negative exactly when the two factors have opposite signs, which on $t \geq 0$ happens for $0 < t < 2$. So the position graph is concave down on $0 < t < 2$ and concave up for $t > 2$; $t = 2$ is an inflection point. *Credit any correct reasoning that reaches the interval $0 < t < 2$ from the sign of $a(t)$ — a sign chart, the factored form, or test values.*